Run report: the summand-swap invariant on the chained S_3 descent

- **Study:** `research/dreg_pair_invariant_20261028/`, pre-registered in `PREREGISTRATION.md` (commit `d708ad2ee`; Amendment 1 after the first secondary run capped, before scoring).
- **Run:** 2026-10-28 (UTC): control 05:51–06:15, primary and soundness 09:30–09:31, secondary 15:21–17:11 (one sparse run, capped).
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.

What ran

- **Tool:** `examples/dreg_invariant.rs` builds three arms from the same committed draw.
  - B: the plain m = 3 chain.
  - Q: adds the swap invariant `p = x₁x₂` over a basis of `span{v_i v_j}`. The first link becomes `s²e₁² + p e₁ + p² + b`, and every equation is quadratic.
  - R: the matched control, adding `r = x₁e₁`. It is also quadratic, but not invariant.
  - `--check` proves each construction by an exact polynomial identity.
- **Draws:** the committed m = 3 draws at (22, 6) (driver study, draws 0–3) and at (21, 7) (scale study: draws 0, 2, 3, 6 unsatisfiable, 1, 4, 5 satisfiable).
- **Engine:** MutantXL closure (`closure_refutation_scan`, dense, at most 130,000 columns). The secondary step used the sparse closure at degree 4, capped at 10^9 entries.
- **Limits:** `prlimit --as=15e9`, `timeout 7200`, one process at a time.
- **Host:** the 4-core, 16 GB container, not isolated. Iso Lab EKS is not reachable from it (no `aws` CLI or credentials). The outcomes are exact algebra; the timings are indicative only.
- **Binary:** `dreg_invariant`, SHA-256 in `runs/binary.sha256`.

Completion

- 30 measurement processes exited 0.
- One secondary run was started under a 10-minute tool cap and stopped by hand; it was relaunched detached.
- One secondary run was stopped by Amendment 1, 39 s after it started.
- Both stopped runs left empty outputs, kept in their own directories.
- Peak RSS: 5.1–5.8 GB (B at degree 4), ≤ 0.13 GB (Q and R at degree 3) and 12.0 GB (the sparse run).

Measured values

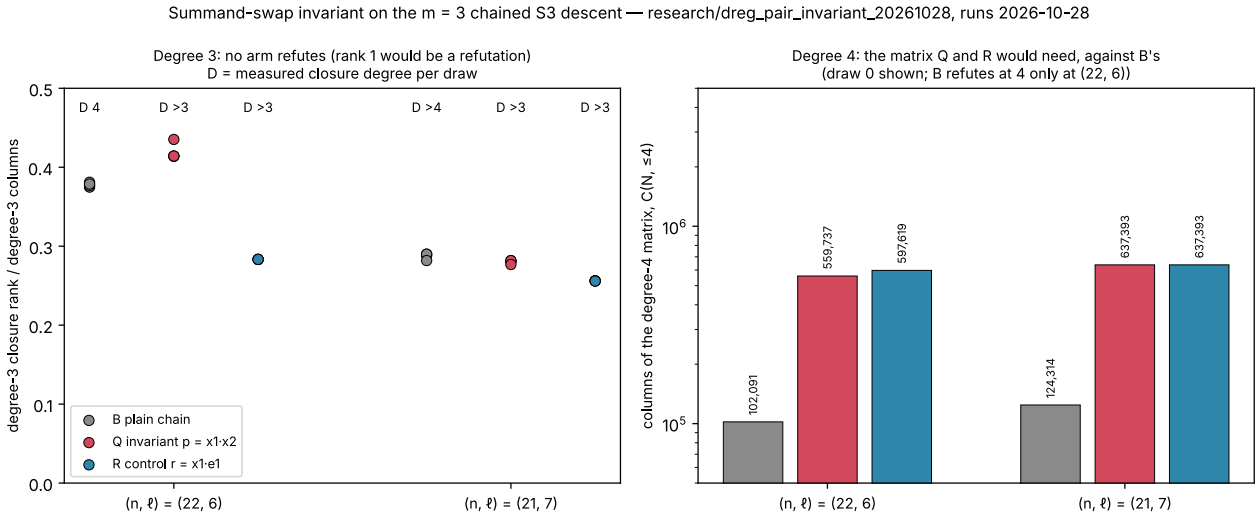


Figure 1: Left: the degree-3 closure rank as a share of the degree-3 columns, one point per draw; no arm refutes at degree 3. Right: columns of the degree-4 Macaulay matrix for each arm (draw 0). Source: `plot.py` (layout only) on `plot-data.json` from `dreg_invariant_score --plot-data`.

- **B:** refutes at degree 4, not 3, on 4 of 4 draws at (22, 6). Saturates degree 4 without refuting on 4 of 4 at (21, 7). This matches the committed records.
- **Q and R:** do not refute at degree 3 on any of the 8 unsatisfiable draws. Degree 4 is unmeasured: the sparse run capped, and the dense matrix needs 523,686–637,393 columns.
- **Validity:** the identity check is true on 22 of 22 systems, the positive control is reproduced, and no satisfiable draw is refuted. Valid.

- **Registered verdicts:** H_none at (22, 6), which is measured, and H_none at (21, 7), which follows the registered convention for an unmeasured degree 4. Q is unmeasured below B's degree on 4 of 4 draws at (21, 7), so a lower degree is not excluded there.
- **Per-run values:** in `score-output.txt` and `runs/`.